

AIM

To write a C program to create a singly linked list and perform operations such as adding elements at the front, adding elements at the back, and printing all the elements in the list without using double pointers.

PROCEDURE

1. Start the program.
2. Define a structure for the linked list node containing data and next pointer.
3. Create a function to allocate memory and create a new node.
4. Create a function to add a node at the front of the linked list.
5. Create a function to add a node at the back of the linked list.
6. Create a function to display all the elements of the linked list.
7. In the main function, call the required functions and update the head pointer.
8. Display the linked list elements.
9. Stop the program.

PROGRAM

```
/*
File Name : list.c
Author    : Monish Kanna
Purpose   : To create a singly linked list and perform operations such as adding elements at the
            front, adding elements at the back, and displaying all the elements in the list.

*/
#include <stdio.h>
#include <stdlib.h>
struct Node
{
    int data;
    struct Node *next;
};
struct Node* createNode(int value)
{
    struct Node *newNode = (struct Node*)malloc(sizeof(struct Node));
    newNode->data = value;
    newNode->next = NULL;
    return newNode;
}
struct Node* addAtFront(struct Node *head, int value)
{
    struct Node *newNode = createNode(value);
    newNode->next = head;
    return newNode;
}
struct Node* addAtBack(struct Node *head, int value)
{
    struct Node *newNode = createNode(value);
    if (head == NULL)
        return newNode;
```

```

    struct Node *temp = head;
    while (temp->next != NULL)
        temp = temp->next;
    temp->next = newNode;
    return head;
}
void printList(struct Node *head)
{
    printf("Linked List elements:\n");
    while (head != NULL)
    {
        printf("%d -> ", head->data);
        head = head->next;
    }
    printf("NULL\n");
}
int main()
{
    struct Node *head = NULL;
    head = addAtFront(head, 10);
    head = addAtFront(head, 20);
    head = addAtBack(head, 30);
    head = addAtBack(head, 40);
    printList(head);
    return 0;
}

```

OUTPUT

```

PROBLEMS  TERMINAL  OUTPUT  PORTS
PS C:\AMI C PROGRAM> cd "c:\AMI C PROGRAM\Monish_kanna_week3_practice1\"
• Linked List elements:
  20 -> 10 -> 30 -> 40 -> NULL
• PS C:\AMI C PROGRAM\Monish_kanna_week3_practice1>

```

RESULT

Thus, the C program to create a singly linked list and perform insertion at the front and back without using double pointers was executed successfully and the linked list elements were displayed correctly.